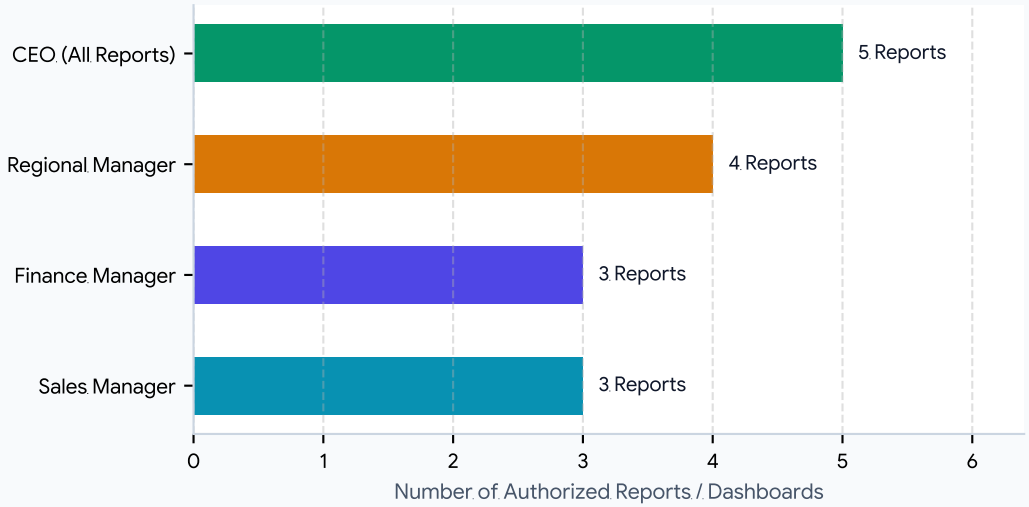
EXPERIMENT 9 — PAGE 1: SECURITY ROLE & ACCESS OVERVIEW (RBAC)

Part A: Role-Based Access Control (RBAC), Report Entitlements & Governance Principles

1. Role-Based Access Control (RBAC) Permission Matrix

Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Sales Manager	Yes	No	No	No
Finance Manager	Yes	No	No	No
Regional Manager	Yes	No	No	No
BI Administrator	Yes	Yes	Yes	Yes

2. Authorized Report Coverage by Role



3. Organizational Role & Information Requirements Matrix

Role	Information Requirements	Reports and Dashboards Access
CEO	Overall sales, profit, orders & business performance	All reports (5 Dashboards)
Sales Manager	Daily sales, orders, products & customer performance	KPI Monitoring, Daily Sales, Monthly Executive
Finance Manager	Profit, profit margin & financial performance	KPI Monitoring, Weekly Profit, Monthly Executive
Regional Manager	Region-wise sales, profit & performance trends	KPI, Monitoring Scenario, Daily Sales, Weekly Profit

LEAST-PRIVILEGE ACCESS POLICY

- CEO: Views all 5 organizational dashboards (read-only executive view).
- Sales Manager: Restricted to KPI, Daily Sales & Monthly Executive.
- Finance Manager: Restricted to KPI, Weekly Profit & Monthly Executive.
- Regional Manager: Restricted to regional KPIs & scenario reports.
- BI Administrator: Retains full Edit, Share, and Workspace Admin rights.

EXPERIMENT 9 — PAGE 2: ROW-LEVEL SECURITY (RLS) DESIGN & DATA ISOLATION

Part B: DAX Row-Level Security Filters, Regional Sales Isolation & Security Benefits

TOTAL RLS ROLES DEFINED

5 Roles

1 Executive + 4 Regional

REGIONAL RESTRICTIONS

4 Territories

West • East • Central • South

EXECUTIVE ACCESS SCOPE

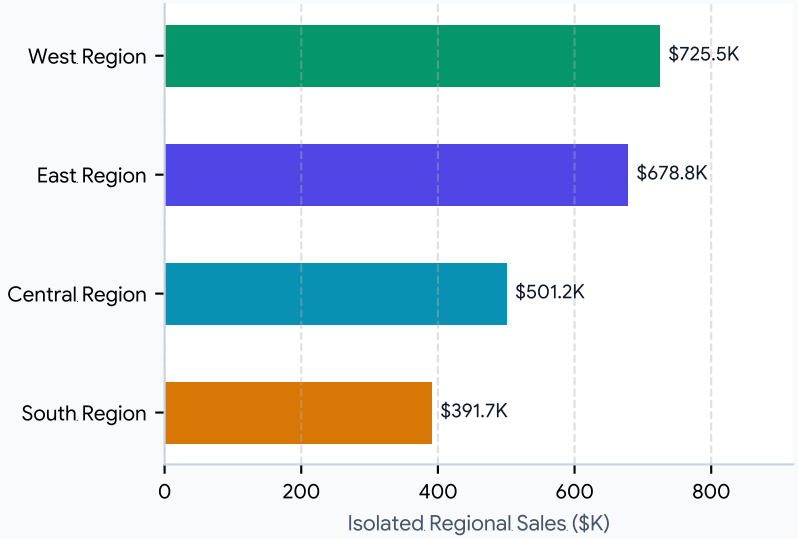
All Regions

$12,642,502 Global Sales

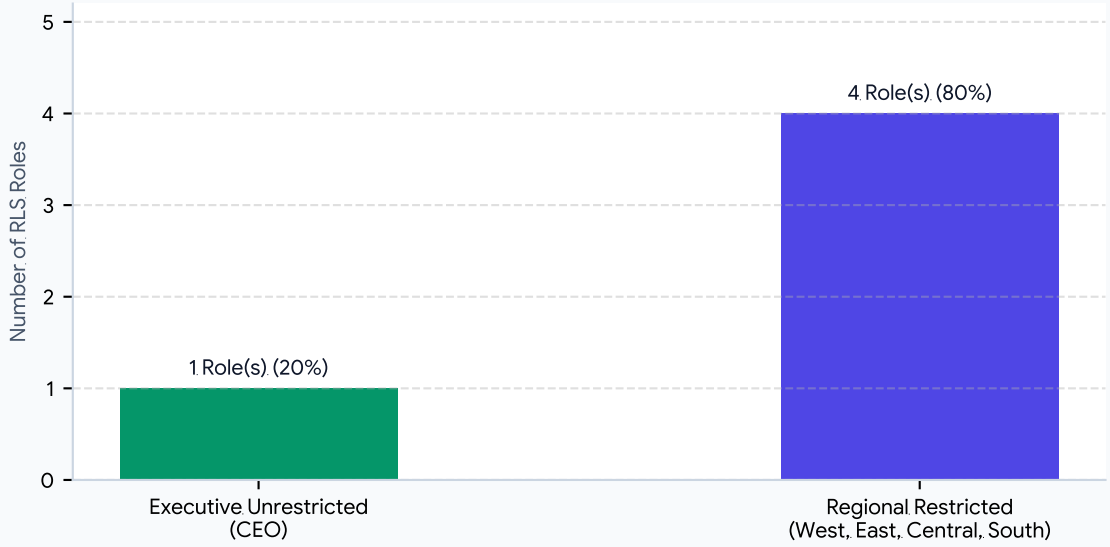
1. Regional Row-Level Security (RLS) DAX Filter Matrix

Role	Assigned Region	Data Access	RLS DAX Filter
CEO	All Regions	All regional data	No regional restriction
Regional Manager - West	West	West region only	[Region] = "Western US"
Regional Manager - East	East	East region only	[Region] = "Eastern US"
Regional Manager - Central	Central	Central region only	[Region] = "Central US"
Regional Manager - South	South	South region only	[Region] = "Southern US"

2. Sales Isolated per US RLS Role



3. RLS Role Classification: Unrestricted vs. Restricted



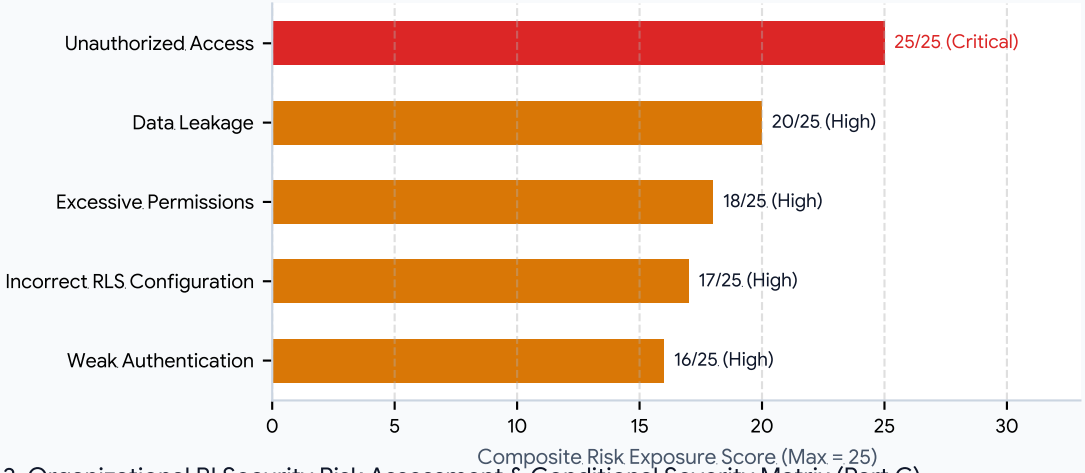
HOW ROW-LEVEL SECURITY (RLS) IMPROVES DATA SECURITY

1. Granular Data Isolation: Restricts regional managers to viewing only rows matching their assigned territory (e.g., West sees only $725.5K).
2. Cross-Territory Breach Prevention: Blocks unauthorized viewing of competitor/peer regional margins without duplicating PBIX files.
3. Confidential Financial Protection: Keeps global profit ($1.47M) and executive metrics visible exclusively to the CEO role.
4. Scalable Single-Report Governance: Enables safe department-wide sharing.

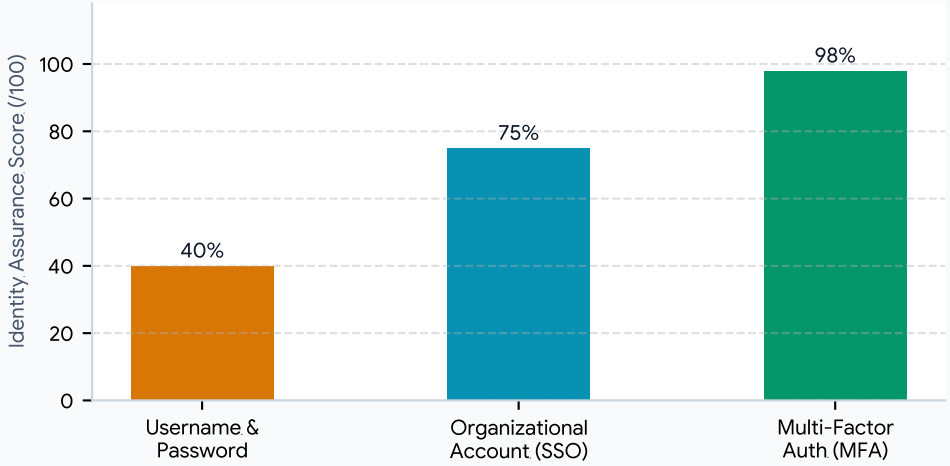
EXPERIMENT 9 — PAGE 3: SECURITY RISK MATRIX, AUTHENTICATION & CONTROLS

Parts C & D: Threat Severity Scoring, Authentication Layer Comparison & Mitigation Action Plan

1. BI Security Risk Severity Ranking (1 Critical, 4 High)



2. Authentication Layer Security Assurance Comparison



3. Organizational BI Security Risk Assessment & Conditional Severity Matrix (Part C)

Security Risk	Likelihood	Impact	Risk Level	Recommended Mitigation Control
Unauthorized Access	High	High	Critical	Implement authentication and MFA
Excessive Permissions	Medium	High	High	Apply least privilege access
Data Leakage	Medium	High	High	Restrict sharing and export permissions
Weak Authentication	Medium	High	High	Enforce strong passwords and MFA
Incorrect RLS Configuration	Medium	High	High	Test and validate RLS roles

4. Security Recommendations Matrix & Expected Outcomes (Part D)

Security Issue	Recommended Control	Priority	Expected Outcome
Unauthorized Access	Implement MFA	Critical	Stronger user verification
Excessive Permissions	Apply Least Privilege Access	High	Restricted unauthorized actions
Data Leakage	Restrict Sharing and Exporting	High	Improved data protection
Weak Authentication	Enforce Strong Authentication	High	Reduced account security risks
Incorrect RLS Configuration	Regularly Test RLS Roles	High	Accurate regional data restrictions